Referential adaptation in the face of changing input frequencies

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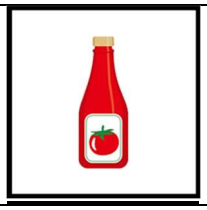
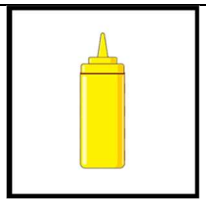
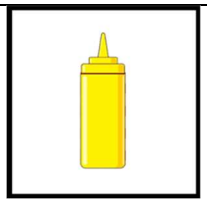
Adaptation to both syntactic and referential patterns is well established (1,6): comprehenders favor recently-encountered patterns during the interpretation of new input. But we don't know how flexible adaptation is: do people rapidly shift to new patterns? We test this for referential adaptation. Ambiguous pronouns (Table 1, top) tend to be assigned to the prior subject (Thomas). Through adaptation, this bias diminishes following exposure to pronouns that unambiguously refer to the nonsubject (bottom panel; 6). But it is unknown how stable this adaptation is. If people suddenly encounter a different reference pattern, do biases shift?

We consider three possibilities: 1) A recency bias; 2) a situation-based primacy bias, and 3) a combination of the prior subject-interpretation bias and the situation bias. The two major theoretical accounts in this area, 1) residual activation and 2) error-based (implicit) learning, make different predictions. Residual activation briefly makes primed structures easier to retrieve, predicting a recency effect (5). Error based learning (EBL) proposes that speakers predict upcoming language and recalibrate their expectations when their predictions are incorrect (2,3), predicting long-lasting effects of adaptation (1), but only (or more strongly) for surprising input (6). This predicts a stronger effect of the infrequent nonsubject-exposure input when it follows vs. precedes a block of the frequent subject-exposure input. A third possibility is that people favor their prior biases, when this approach is tenable, and utilize a heightened attention to the initial pattern in the experiment to decide whether their current situation aligns with their lifetime experience. This primacy effect has not been shown for adaptation studies, but is consistent with situation- or speaker-specific effects elsewhere in the literature (4, 7).

Methods: We test whether adult English native speakers can learn from exposure to two distinct referential patterns in a single experimental session. **Exp. 1** (MTurk; N=74) subjects read 44 stories about 4 characters. Critical questions probed pronoun interpretation. Exposure stories unambiguously referred to the subject or nonsubject character. Subjects were exposed to two blocks of stimuli (Fig. 1), half (List 1) with subject-references then nonsubject-references, the other half in reverse order. Each block started with 4 exposure items to initiate the priming effect. The rest of the block contained a mix of 8 critical test items and 10 more exposure items. Ambiguous critical items were distributed throughout each block to measure the effect of exposure; the outcome variable was whether the subject or nonsubject character was chosen as the referent (see Table 1). **Exp. 2** (Prolific; N=100) repeated the blocked priming procedure from exp. 1 using audio stimuli. Each block featured different speakers (Block 1-Female; Block 2-Male) and different sets of 4 characters to enhance the sense of two discourse contexts with distinct speakers and topics**. Results & Discussion:** In both experiments, subjects adapted in the first block, assigning the pronoun to the subject more in the subject- than nonsubject-exposure condition (Figs. 2&3). But critically, they shifted their response bias for block 2 only in List 2, with the less common nonsubject pattern first. Results suggest that participants follow the initial pattern to establish a situation-specific expectation. But for the infrequent pattern, this bias is fragile, and they quickly return to their baseline subject bias when new input is consistent. This is inconsistent with both the residual activation and EBL accounts. The lack of adaptation observed in the second, nonsubject exposure block creates a serious challenge for the predictions of EBL because prediction error should be high in this condition and relatively low when subject exposure occurs in the second block. Results suggest that adaptation reflects situation-specific tuning that is constrained by prior biases.

Table 1- Example Stories with Answer Choices

Critical Story: Thomas and David were going to a concert. Thomas offered the tickets to David, and then he waited in line. (Who waited in line?)			
Exposure Story: Ana and Will were having lunch at McDonald's. Ana ate french fries with Will and then [she/he] spilled ketchup on the table.			

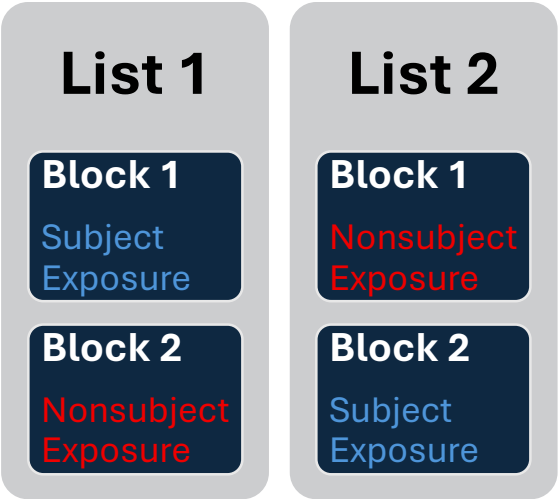


***Fig. 1**-Block structure in* Exps 1&2.

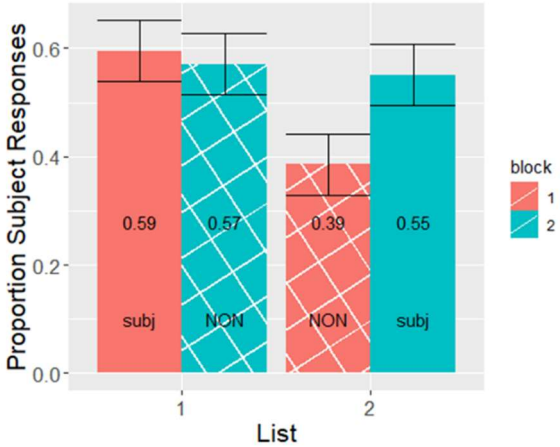


***Fig. 2**-Subject responses in* Exp. 1 *(critical interaction: β=1.25, SE=.42, z=2.96, p<.01)*

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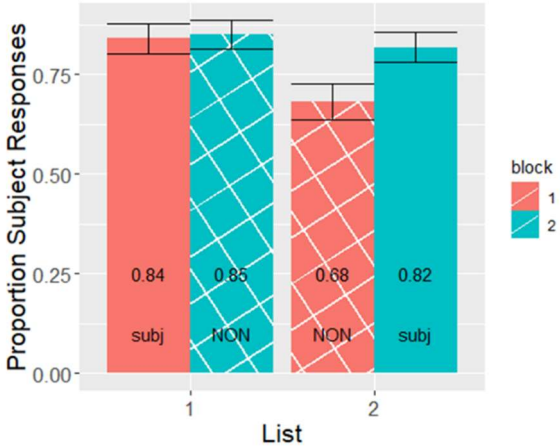

***Fig. 3**-Subject responses in* Exp. 2 *(critical interaction: β=.94, SE=.35, z=2.7, p<.01)*